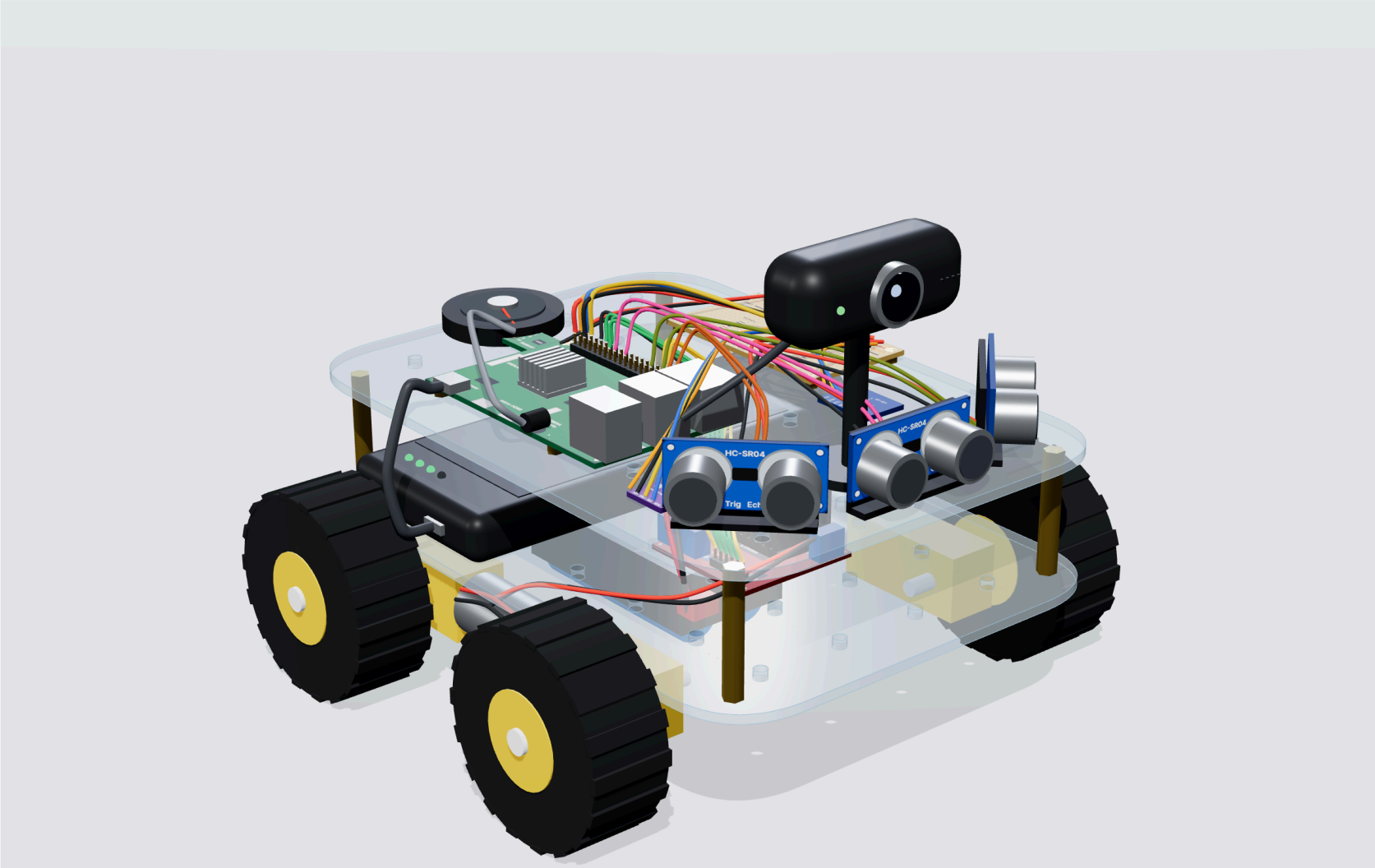
RAQIB · THE FINISHED ROBOT · 3D VIEW

The finished robot

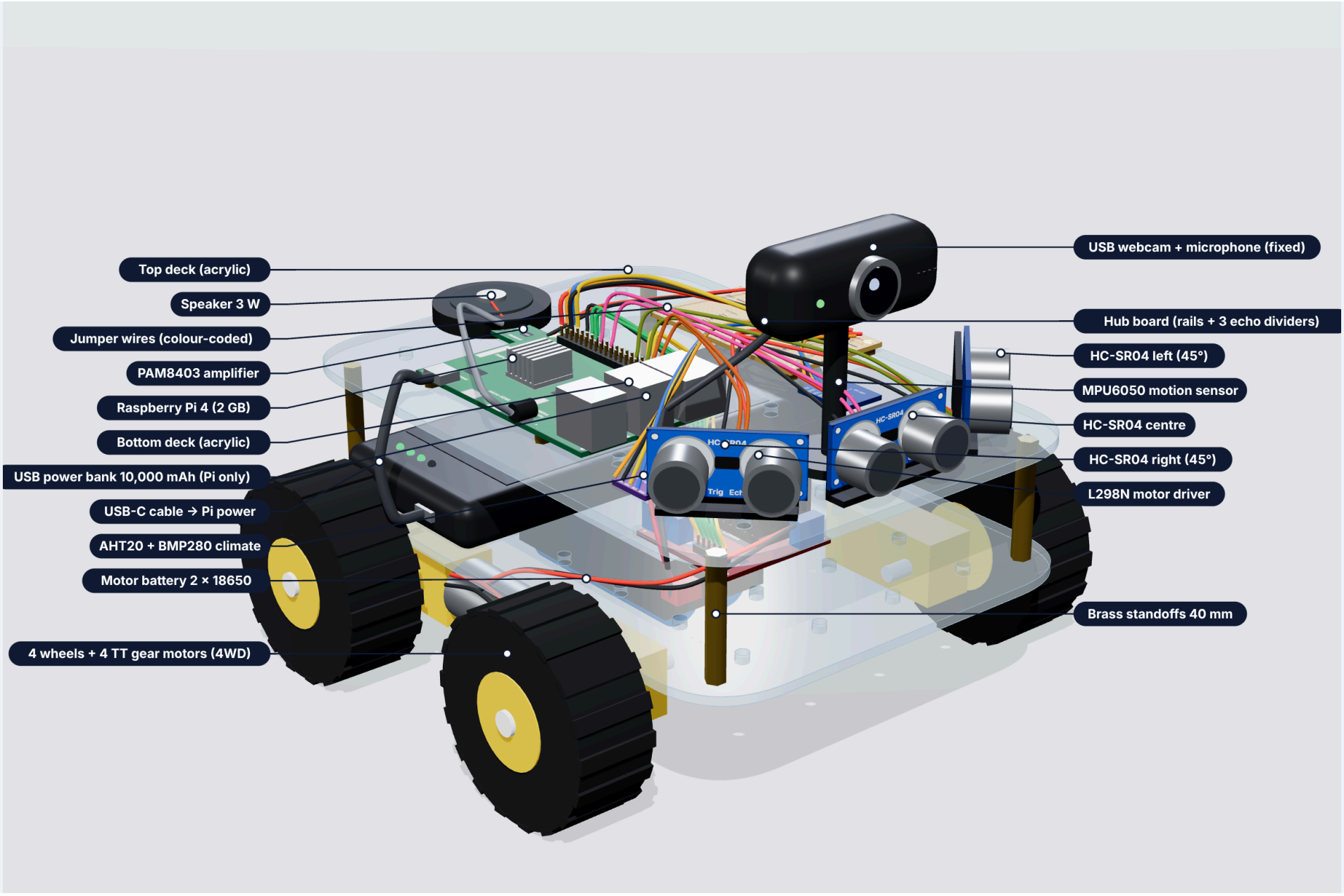
Raqib 2.5, the reliable budget AI inspection robot, fully built and wired: four-wheel drive, two acrylic decks, three ultrasonic sensors and a USB webcam at the front, the Raspberry Pi and sensors on top, the power bank, motor driver and motors below.



A 3D model at real size (mm), built from the parts and layout of the build guide. Board colours and printed labels follow the common versions of each part; your boards may differ slightly.

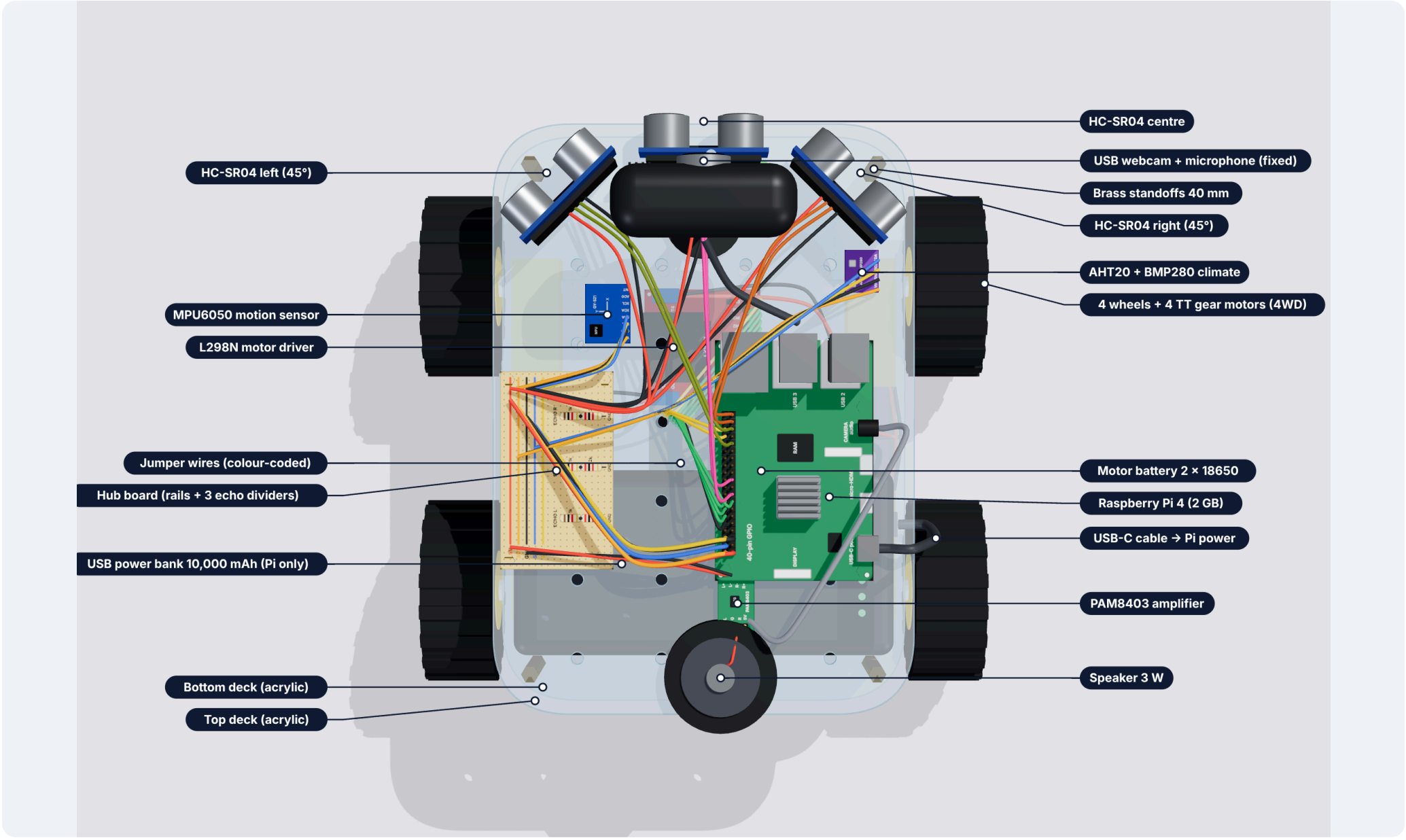
Every part, labelled

All parts in their final place. Front of the robot = the side with the webcam.



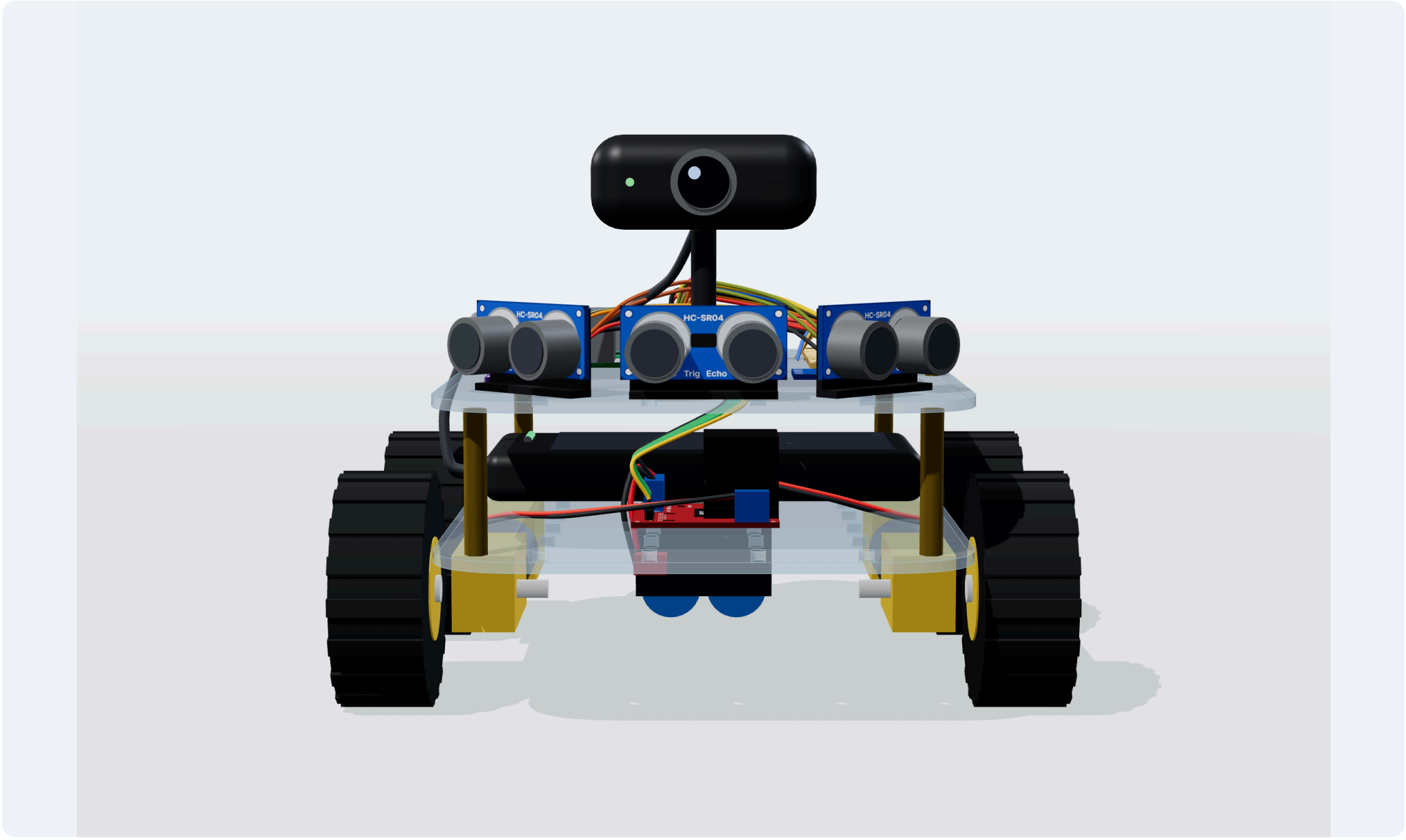
Seen from above: how the top deck is organised

Front (top of the picture): three HC-SR04 fanned left / centre / right and the webcam on its post. Middle: Raspberry Pi with its 40-pin header facing the hub board, MPU6050 near the centre, AHT20 at the edge away from the Pi. Back: PAM8403 and the speaker.



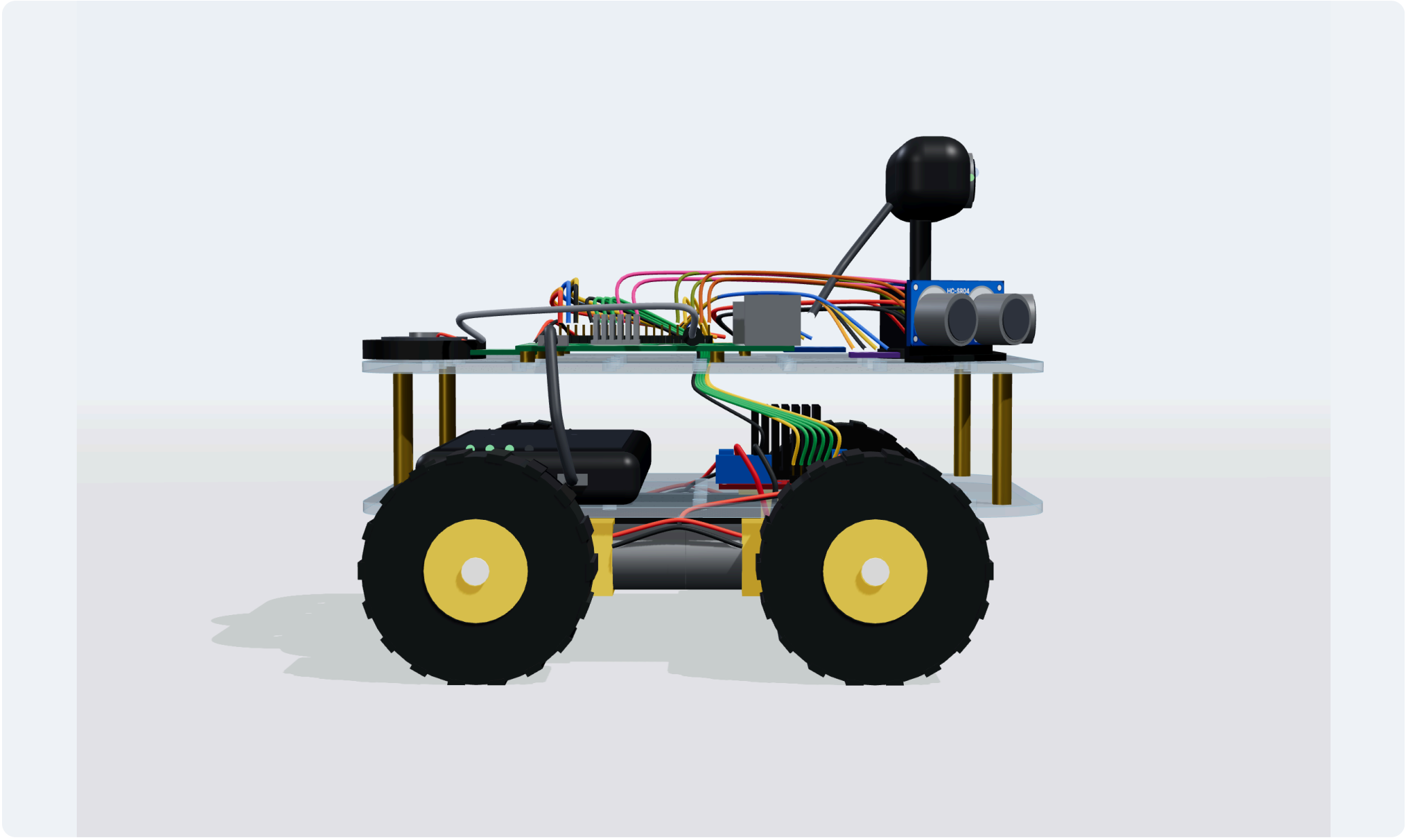
Front view

What the robot looks at: the centre sonar straight ahead, the left and right sonars turned 45°, the webcam above them. The camera is fixed: the robot turns its whole body to look around.



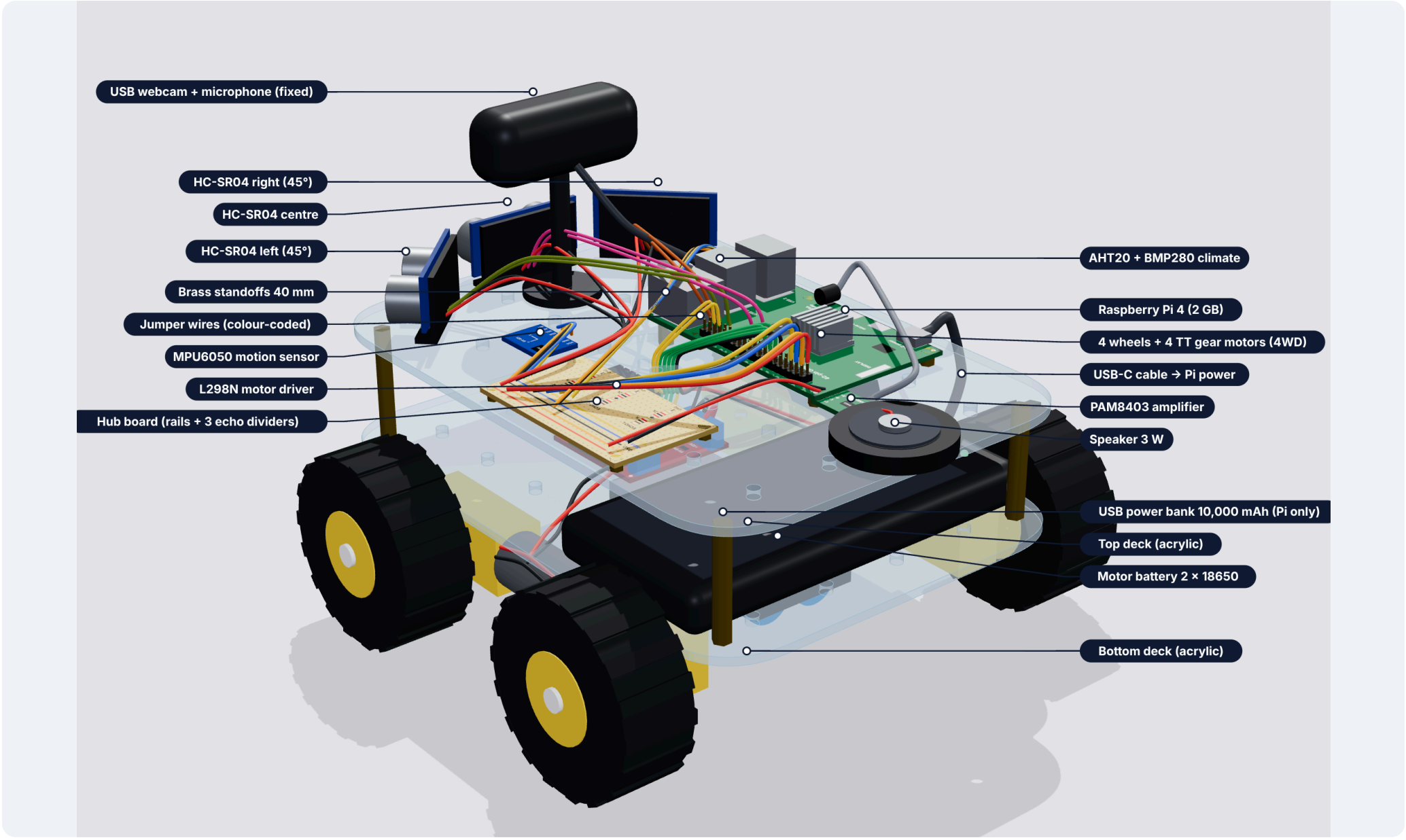
Side view

Two decks on 40 mm brass standoffs. Bottom deck: the USB power bank (the Pi's own battery) and the L298N motor driver; under it the 2 × 18650 motor battery and the four TT motors with their wheels (4WD). Top deck: Pi, sensors, speaker, webcam.



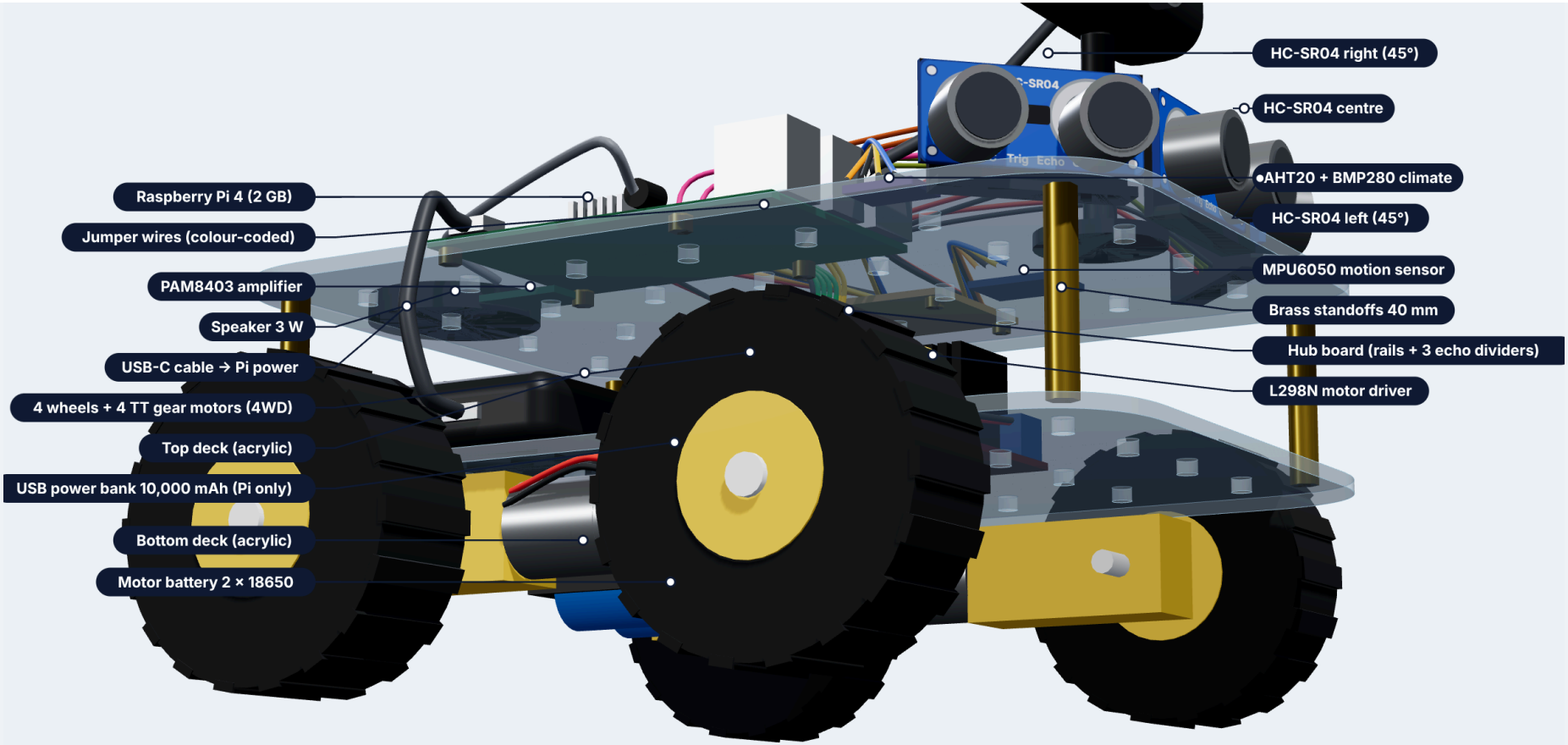
From the back

The power bank's own USB-C cable goes round the deck edge to the Pi's power socket; the speaker and PAM8403 sit at the back; the wires from the Pi go down through the deck to the L298N, including pin 9, the common ground.



Underneath

The 2 × 18650 holder hangs under the bottom deck with its switch reachable and powers only the motors; four TT motors, two per side, each pair wired together to one side of the L298N. No encoders: the robot times its motors.



Close-up of the top deck

Colour-coded jumper wires from the Pi's 40-pin header: red 5 V, orange 3.3 V, black GND, blue SDA, yellow SCL, green motor direction, amber motor speed, pink / olive / brown sonars.

